Campus Verse – Product Requirements Document (PRD)

1. Executive Summary

Campus Verse is an advanced, multi-institution ERP platform designed to digitize and centralize operations for schools, colleges, and universities, including multi-branch and multi-city networks. It enables institution groups to manage academic, administrative, financial, and operational processes in a unified way while preserving the autonomy and configurability of each branch and institution type.

The core value proposition is to streamline institutional operations, improve efficiency, and empower education groups with real-time, cross-institution visibility through a single owner dashboard and role-based control across all branches, states, and cities.

2. Product Overview



Campus Verse supports three primary institution archetypes, each with its own data architecture and hierarchy model:

- Schools (multi-state, multi-city, multi-branch; class-section structure)
- Colleges (e.g., grades 11–12; stream-based structure such as MPC, BiPC, CEC)
- Universities (department and year-based structure such as ECE, CSE, EEE with year/section)

All institution types can coexist under a single owner account, allowing an education group that runs schools, colleges, and universities to see unified metrics (students, staff, finance, attendance, etc.) while being able to drill down by institution type, state, city, branch, department, class, year, and section.

2.1 Product Vision

To become the standard operating system for multi-institution education groups, enabling data-driven decisions, operational excellence, and seamless digital experiences for owners, admins, staff, students, and parents.

2.2 Target Customers

- School groups operating multiple branches across states and cities
- Intermediate / junior colleges offering 11th–12th grade in multiple streams
- Universities with multiple departments, years, and sections
- Single institutions planning to scale into multi-branch networks

2.3 Key Personas

- **Group Owner / Chairman**– Needs unified dashboards and high-level control.
- **Institution Head (Principal / Dean / Director)** – Needs institution-level operational control and reports.
- **Branch Admin** – Manages day-to-day operations at a branch.
- **HOD / Coordinator**– Manages departments, classes, or years.
- **Teachers / Faculty**– Manages academics, assessments, and communication.
- **Accountant / Finance Officer**– Manages fees, collections, dues, and expenses.
- **HR & Payroll Officer**– Manages staffing, attendance, and payroll setup.
- **Transport / Hostel / Infra Managers**– Manages logistics and facilities.
- **Students & Parents**– Consume information, pay fees, view reports and communication. [Assumption: dedicated portals or app]

2.4 High-Level Goals and Success Metrics

- Reduce manual administrative workload by at least 40 percent within 6 months of deployment.
- Achieve 95 percent data completeness for student, staff, and financial records across branches.
- Enable owners to get real-time, unified KPIs in under 10 seconds on the dashboard.
- Ensure fee collection tracking accuracy above 99 percent across all branches.

- Reach 80 percent monthly active usage among staff for core modules (Attendance, Academics, Exams, Finance, HR).

3. System Architecture Overview



Campus Verse uses a hierarchical, multi-tenant architecture to model institution groups, institutions, and branches while maintaining secure data segregation and flexible configuration.

3.1 Hierarchy Model

Global Owner Level

- Owner can manage multiple institution types (School, College, University) under a single group.
- Unified dashboard for high-level metrics: total students, staff, fee collections, dues, transport usage, hostel occupancy, etc.
- Cross-institution filters: by institution type, state, city, branch, academic year.

Geographical Hierarchy

- State → City → Branch structure for all institution types.
- Owner or super admin can create:States (e.g., Telangana, Andhra Pradesh),Cities within state (e.g., Hyderabad, Vizag),Branches within city (e.g., XYZ School – Kukatpally).

Institution-Type-Specific Academic Hierarchies

Schools

- Branch → Classes (e.g., Grade 1 to 10) → Sections (A, B, C, D) → Students
- Each branch can define its own class and section structure with branch-level settings.

Colleges (11th–12th)

- Branch → Streams (e.g., MPC, BiPC, CEC) → Classes/Batches (e.g., MPC-A, MPC-B, IIT batch, NEET batch) → Students
- Stream-specific timetables, fee structures, and academic rules.

Universities

- Branch → Departments (e.g., ECE, CSE, EEE) → Years (1st, 2nd, 3rd, 4th year) → Sections → Students
- Supports departmental heads, program coordinators, and complex exam structures.

3.2 Multi-Tenancy and Data Isolation

[Assumption: Logical multi-tenancy per institution group on a shared infrastructure.]

- Each owner group is a tenant; data is isolated at tenant level.
- Branches share configurations within a tenant but can override at branch level where allowed (fee rules, holidays, communication templates, etc.).
- Role-based access control (RBAC) enforced at tenant, institution, branch, department, and class levels.

3.3 Integration and Extensibility

[Assumption: REST APIs available for key modules.]

- Payment gateways for online fee collection.
- SMS gateways with DLT compliance.
- Optional: Email providers, LMS tools, video conferencing tools for online classes, accounting tools. [Assumption]

4. User Journeys



4.1 Owner – Setting Up a New School Network

- Create institution type: School XYZ Group.
- Define states, cities, and branches where schools operate.
- For each branch: configure academic batches, classes, and sections.
- Use Organization module to create branch admins and assign roles.
- Bulk upload student and staff data per branch.
- Configure fee structures, transport, and hostel (if applicable).
- Review unified dashboard for total students, fee projections, and staffing.

4.2 College Owner – Setting Up Streams

- Select institution type: College.
- Create state → city → branch.
- Within branch, define academic streams: MPC, BiPC, CEC, etc.
- Create classes/batches under each stream (e.g., MPC–Regular, MPC–IIT Batch).
- Upload students mapped to stream and class.
- Configure stream-wise timetables, fee plans, and exams.

4.3 University Admin – Department and Year Structure

- Create university branches (e.g., ABC University – Hyderabad).
- Create departments: ECE, CSE, EEE, etc.
- Create year-wise entities: 1st, 2nd, 3rd, 4th year for each department.
- Under each year, define sections and upload student data.
- Assign HODs and year coordinators via Organization and HR modules.

5. Core Platform Requirements

5.1 Organization Module



Objective: Centrally manage institutional hierarchy, roles, permissions, and foundational settings across branches and institution types.

5.1.1 Features

- **Academic Batches**– Create and manage academic years/terms (e.g., 2024–25) with start and end dates.
- **Classes** – Define class structures for schools (grades), colleges (stream-specific classes), universities (year/section structures).
- **Subjects** – Create subject catalogs per institution type and map to classes/streams/departments.
- **User Management**– Create, edit, and deactivate users (staff, admins, owner delegates).
- **Role Management** – Define roles (e.g., Group Admin, Branch Admin, Accountant, HOD, Teacher) with RBAC permissions per module.
- **Branches** – Manage branch metadata (location, contact, institution type, infrastructure tags).
- **Org/Branch Logos**– Upload and manage branding used in portals, receipts, ID cards, and reports.
- **Branch-Level Settings**– Fine-tune tax rates, fee rules, communication settings, holidays, weekly offs at branch level.

- **Org Information**– Group-level information: legal entity name, address, contact, regulatory identifiers. [Assumption]
- **Schedule Jobs**– Configure background jobs (daily attendance summaries, fee reminders, report generation).
- **Undo Approvals**– Allow authorized roles to revert certain approvals (fees, leaves, certificates) with audit logs.
- **Approval Email Settings**– Configure email notifications for approval workflows.
- **Custom Form Settings**– Build and manage custom data capture forms used across modules.
- **Sort Settings** – Configure ordering for lists (students, classes, branches) by default parameters.

5.1.2 Requirements and Constraints

- Must support at least 5,000 concurrent users across an owner group. [Assumption]
- All role and branch changes must be auditable with timestamp, actor, and action details.
- Hierarchy changes (e.g., class re-structuring) must not corrupt historical academic and finance records.

5.2 Finance Module



Objective: Provide end-to-end fee lifecycle management, including fee setup, collection, concessions, cancellations, refunds, and reporting across all branches.

5.2.1 Features

Fee Counter

- Make Payment – Record student fee payments (cash, card, online, cheque, wallet).
- Fee Bulk Upload – Bulk upload fee data for multiple students or academic sessions.
- Bulk Upload – General bulk operations for fee assignments or corrections. [Clarify overlap in design stage]
- Concession Bulk Upload – Bulk apply concessions for eligible students (scholarships, staff wards, etc.).

Fee Management

- Assign Fee to Students – Map fee structures to students based on class, stream, department, transport/hostel usage, and scholarships.
- Yet to Assign Fee Reports – Identify students and classes with missing fee assignments.

Paid & Due Reports

- Total Paid & Due Reports – Branch-wise and group-wise overview of fees collected vs outstanding.
- Class Wise Due Reports – Dues per class/section.
- Student Wise Due Reports – Detailed dues and payment history by student.
- Branch Wise Due Reports – Compare performance across branches.
- Sibling Due Reports – Track dues at family level for siblings.
- Installment Wise Student Fee Due Reports – Track dues by installment schedule.

Collection Reports

- Fee Day Sheet – Daily summary of collections by mode and collector.
- Fee Type Report – Collections by fee type (tuition, transport, hostel, exam, etc.).
- Student Fee Statement – Consolidated fee history for a student.
- Fee Type Wise Transactions – Transaction list filtered by fee type.
- Class Wise Concession Report – Concession values per class/section.
- Student Wise Concession Report – Concessions per student.

Fee Setup

- Bank Settings – Configure bank accounts and reconciliation fields.
- Branch Method Settings – Define allowed payment methods per branch.
- Fee Receipt Number Settings – Pattern and sequencing per branch or institution.
- Org Fee Settings – Global rules: due dates, penalties, automatic reminders.
- Fine Settings – Late fee calculations (fixed/percentage, grace periods).
- Concession Settings – Eligibility criteria, approval workflows, limits.
- Cancellation Settings – Rules for fee cancellation and reversals.

- Application Fee Settings – Non-refundable application/registration fee configurations.
- Wallet Categories & Wallet Sequence Settings – Internal wallet management for advances, deposits, and adjustments.
- Payment Mandatory Field Settings – Configurable fields required during collection (e.g., reference no., remark).

Approval

- Fee Cancellation Request – Workflow for approving fee cancellations.
- Student Fee Refund Requests – Track and approve refunds with audit logs.
- Concession Requests – Approval flows for new or adjusted concessions.

5.2.2 Performance and Compliance

- All financial operations must be ACID-compliant. [Assumption: backed by transactional DB]
- Must retain detailed transaction history for at least 7–10 years, configurable per region. [Assumption]
- Should support export of data to accounting systems via CSV/API.

5.3 Admissions Module



Objective: Manage the full admission lifecycle: leads, applications, admissions, alumni, and agent tracking, ensuring no follow-up is missed.

5.3.1 Features

Enrollments

- Applications – Capture and manage student applications per branch and institution type.
- Admissions – Convert applications to confirmed admissions with fee linkage.
- Alumni Enrollment – Tag students as alumni and maintain basic records.
- Leads – Manage enquiries from agents, walk-ins, online forms, and campaigns.

Settings

- Application Number Settings – Patterns, prefixes, per branch or per institution.
- Admission Number Settings – Unique admission number sequences.
- Enrollment Code Settings – Enrollment codes for internal reference.
- Admission Fields Setting – Customizable fields for admission forms.
- Enquiry Stages – Custom lead stage pipeline (New, In Progress, Converted, Lost).
- Organization Sources – Source taxonomy (Agents, Social Media, Website, Walk-in).
- Leads Code Settings – Codes or tags for different lead categories.
- Task Number Settings – For follow-up task management.
- Organization Streams – Stream setup for colleges (MPC, BiPC, CEC, etc.).

Followups

- Leads Followup – Assign, schedule, and track follow-up tasks, with reminders and outcomes.

Bulk Upload

- Bulk Admissions – Upload confirmed admission data from external systems or spreadsheets.

Web Forms

- Dynamic Forms – Build online forms for enquiries and applications integrated into websites.

Reports

- Leads Report – Source, stage, and conversion analytics.
- Application Report – Applications by branch, class/stream, and status.

Agent Tracking

- Live Tracking – Track leads, applications, and conversions associated with agents; monitor performance.

5.4 H.R Module



Objective: Manage the entire staff lifecycle, including enrollment, attendance, leave, workload, and approvals.

5.4.1 Features

Staff Enrollment

- Enroll Staff – Add new staff with roles, branches, and departments.
- Staff Bulk Enroll – Import staff lists in bulk.
- Staff Information Updation – Update profile, contact, role, and employment details.

Staff Attendance

- Daily Attendance – Mark presence/absence at branch or department level.
- Daily Attendance Report – Day-wise attendance summary.
- My Attendance – Self-view for staff members.
- Mark Staff Attendance – Manual marking for authorized users.
- Attendance Summary – Monthly/period attendance metrics.

Staff Leave

- Apply Leave – Staff can submit leave requests.
- Staff Leave History – Personal leave history and balances.

Reports

- Staff Attendance Report – Filterable by branch, department, role, and date range.
- Staff Details Report – Active and historical staff information.
- Staff Leaves Report – Summaries of leaves by type and duration.
- Inactive Staff Details – Records of resigned or terminated staff.

Settings

- Staff Enrollment Sequence Number Settings – ID/employee number formats.
- Shifts – Configure working shifts and timings.
- Leave Management – Leave types, entitlements, carry-forward rules.
- Holidays – Shared and branch-specific holiday calendars.
- Weekly Offs – Weekly off-day configuration by role or branch.
- Assign To Branches – Allocate staff to one or more branches.

Departments and Designations

- Create/View Departments – Academic and non-academic departments.
- Create/View Designations – Positions and roles within the organization.

Staff Workload

- Allocate Workload – Assign teaching loads, subjects, and duties.
- Teacher Workload – Track teaching hours and responsibilities.

Approvals

- Staff Leave Approval – Approve/reject staff leave requests.
- Staff Resignation Approval – Manage exit approvals and termination workflows.

5.5 Communications Module



Objective: Centralize communication across SMS, announcements, and event calendars to ensure timely and targeted messaging.

5.5.1 Features

SMS

- Send SMS – Send bulk or targeted SMS to students, parents, staff, and contacts.

Approvals

- Approve Announcements – Workflow for approving major announcements.
- Announcement Approvals – Track approval status and history.

Event Calendar

- Event Types Create/View – Define event categories (Academic, Cultural, Exam, PTM).
- Event Create/View – Create events with date, time, audience, and reminders.

Announcements

- Announcements – Publish texts/notices to target groups.
- Announcement Settings – Configure visibility rules and channels.

Settings

- Communication Templates – Template library for recurring messages.
- DLT Template – Manage DLT-approved templates and IDs.
- Credentials – SMS/communication gateway credentials.
- Password Generation – Configure password policies and communication for user credentials.

Reports

- Sent SMS – Logs of all SMS messages with status.
- Login Reports – User login activity analytics.
- Announcement Reports – Delivered and viewed counts where available. [Assumption]
- Notification Reports – Summary of all notifications sent via integrated channels.
- Never Logged Reports – Identify users who have never logged in.

5.6 Attendance Module

Objective: Track and report attendance for students and optionally staff (complementing HR attendance).

5.6.1 Features

- Daily Attendance – Mark and view daily student attendance per class/section.
- View Attendance – Class-wise and student-wise views.
- Attendance Summary – Consolidated summaries over chosen period.
- Attendance Reports – Exportable reports for compliance and analytics.
- Settings: Holidays – Manage academic holidays impacting attendance calculations.

5.7 Certificates Module

Objective: Issue, approve, and track academic and administrative certificates for students and staff.

5.7.1 Features

- Student Certificates – Issue Certificate and Reports for student-related certificates (bonafide, TC, conduct, etc.).
- Settings – Certificate Sequence Number Settings, Certificate Approval Settings, ID Card Template Settings.
- Approvals – Certificate Approvals and Staff Certificate Approvals for controlled issuance.
- Staff Certificates – Issue Certificate and Reports for employment and experience certificates.
- ID Cards – Generate Student ID Cards and Staff ID Cards using custom templates.

5.8 Student Info Module

Objective: Provide a single source of truth for student data and lifecycle actions.

5.8.1 Features

- Student Details – Full profile including academic and personal details.
- Student Information Updation – Controlled updates of critical data with audit logging.
- Shuffle Student – Move students between sections/classes/streams as allowed.
- Promote Student – Perform academic promotions at batch-end with configurable rules.
- Branch Shifts – Transfer students across branches with history preservation.
- Reports – Student Reports (filtered), Active Student reports for roll counts and capacity planning.

5.9 Academics Module



Objective: Manage class structures, timetables, diaries, PTM scheduling, and academic activities.

5.9.1 Features

- Class Management – Teacher Management and Student Wise Subject Mapping to ensure correct teaching allocation and subject enrollment.
- Time Table – Create/View Time Table for classes, streams, years, and sections.
- Diary – Student Diary, General Diary, Subject Diary Report, General Diary Report to track daily academic and communication notes.
- Approvals – Diary Approvals and Activity Approvals to maintain content quality and compliance.
- Settings – Diary Approval Settings, Activity Settings, Diary Creation Settings.
- PTM – PTM Schedule, PTM Schedule Report for parent-teacher meetings.
- Activities – Activity Creation for academic and co-curricular activities.
- Reports – Subject Dairy Submission Report to monitor academic record-keeping by teachers.

5.10 Transport Module



Objective: Centralize transport operations, including vehicle details, routing, fees, mapping, and tracking.

5.10.1 Features

- Transport Management – Vehicle Details, Vehicle Routes, Route Fee Creation, Student Route Mapping, Trips, Vehicle Tracking, Transport Fee Bulk Upload.
- Reports – Assigned Routes Report to view students and staff per route.
- Settings – Trip Notification Settings to notify parents on trip events. [Assumption: SMS/push integration]

5.11 Expense Management Module

Objective: Track campus operational expenses and provide transparency and basic financial control.

5.11.1 Features

- Expense Tracking – All Expenses, Ledger Book, Expense Transactions.
- Settings – Expense Categories, Expense Voucher Settings.

5.12 Exams & Gradebooks Module



Objective: Support comprehensive assessment management, grades, report cards, and hall tickets.

5.12.1 Features

- Grade Book – Grade Groups, Marks Allocation, Marks Approval, Report Card, Report Card Template, Attributes Allocation, Health and Hygienic Allocation.
- Exams – Exam Structure, Sub Skills, Exam Rule Setup, Schedule Exam, Attributes, Health and Hygienic Details.
- Settings – Hall Ticket Settings, Marks Approval Settings.
- General Exams – Schedule Exam, Marks Entry, Marks Approval, View Marks.
- Bulk Upload – Attendance Bulk Upload for exam attendance records.

5.13 Inventory Module

Objective: Manage educational and administrative inventory including stock, sales, and approvals.

5.13.1 Features

- Inventory Creation – Category Creation, Item Creation, Stores.
- Stock Management – Kit management, Stock Addition, Item Sale, Item Sale Report, Return/Cancellation Requests.
- Approvals – Return/Cancellation Approvals.
- Settings – Inventory Receipt Number Settings, Inventory Discount Settings.
- Reports – Class Wise Item Sale Report, Purchase Report, Student Wise Item Sale Report.

5.14 Infra Management Module

Objective: Manage campus buildings, rooms, and space allocation for general use and exams.

5.14.1 Features

- Building Details – Add Building Details, Add Room Management General, Add Room Management Exams.

5.15 H.R. Payroll Module

Objective: Configure payroll structure and penalty policies for campus staff.

5.15.1 Features

- Payroll Setup – Payroll configuration (earnings, deductions) and Penalization Policy (late marks, absenteeism impact).

5.16 Visitor Management Module

Objective: Securely manage visits to the campus with records and approvals.

5.16.1 Features

- Visitor Management – Visitor Details Setup, Visitors Record (check-in/check-out).
- Visitor Pass – Staff Pass, Student Pass for controlled internal movement. [Assumption: optional]
- Reports – Visitors Report with filters by date, purpose, host.
- Approvals – Visitor Approval for high-security visits.
- Settings – Visitor Approval Settings (rules, workflows).

5.17 Concerns & Feedback Module

Objective: Capture and manage concerns, complaints, and feedback from stakeholders.

5.17.1 Features

- Concerns – Concern Types configuration and Concern Form for submission, tracking, and resolution SLAs. [Assumption: SLA feature in detailed design]

5.18 Hostel Management Module



Objective: Manage hostel infrastructure, occupancy, and kitchen operations.

5.18.1 Features

- Hostel Setup – Hostel Creation, Building Details, Assign Hostels to Branch, Room Management.
- Kitchen Inventory – Item Creation, Purchase Entry, Usage Entry, Meal Plan.
- Reports – Purchase & Usage Report for hostel kitchens.

5.19 Social Post Module

Objective: Coordinate institutional social media presence from a single dashboard.

5.19.1 Features

- Media Setup – Social Media Link configuration (Facebook, Instagram, etc.).
- Post – Create and manage social posts, with scheduling where supported. [Assumption: scheduler]

5.20 Online Class Module

Objective: Facilitate online classes and track attendance.

5.20.1 Features

- Online Classes – Schedule Online Class with subject, teacher, class/section, and virtual meeting link.
- Reports – Online Class Attendance Report.

5.21 Assessments Module

Objective: Design assessments and build a reusable question bank aligned with institutional taxonomy.

5.21.1 Features

- Assessment Setup – Taxonomy, Difficulty Level, My Assessment, Assessment Library.
- Question Setup – Question Bank with filters by subject, topic, level, and taxonomy.

5.22 Lesson Delivery Module

Objective: Plan lessons and track delivery progress against syllabus.

5.22.1 Features

- Lesson Setup – Lesson Creation, Part/Term Settings to structure syllabus.
- Lesson Planning – Lesson Delivery tracking for each class and subject.

6. Unified Owner Dashboard and Analytics



Objective: Provide group-level and branch-level visibility across all institution types on a single pane of glass.

6.1 Dashboard Capabilities

- Unified KPIs: total students, active staff, total fees billed, collected, dues, attendance rates, transport and hostel utilization.

- Filters: institution type (School, College, University), state, city, branch, academic year.
- Drill-down: from high-level metrics to lists (e.g., from total dues to branch-wise dues, then class-wise, then student-wise).
- Alerts: configurable alerts for thresholds (e.g., dues above target, low attendance, underutilized sections).

7. Non-Functional Requirements

7.1 Security and Compliance

- Role-based access control across all modules and hierarchies.
- Audit trails for critical actions (approvals, financial changes, data updates).
- Encrypted storage of sensitive fields where applicable. [Assumption]
- Compliance with local data protection and DLT regulations for SMS.

7.2 Performance and Scalability

- Support tens of thousands of students and staff across multi-branch groups.
- Standard list views must load within 3 seconds for up to 10,000 records with pagination. [Assumption target]
- Dashboard aggregations must refresh in under 10 seconds for owner-level views.

7.3 Availability and Reliability

- Target uptime: 99.5 percent or higher monthly. [Assumption]
- Regular backups with point-in-time recovery options for critical data.

7.4 Usability

- Consistent UI patterns across modules, especially for filters, tables, and bulk actions.
- Localization-ready for multiple languages in future phases. [Assumption]

8. Dependencies, Assumptions, and Open Questions

8.1 Key Dependencies

- Third-party SMS gateways with DLT compliance.
- Payment gateways for online fee collection.
- Video conferencing tools for Online Class module integrations. [Assumption]

8.2 Explicit Assumptions

- Student and parent-facing apps/portals are part of the broader solution but may be delivered in phases.
- APIs will be provided for basic integrations with external tools (LMS, accounting, etc.).
- Institution groups are willing to standardize some processes (e.g., fee structures) at group level where needed.

8.3 Open Questions for Stakeholders

- What regulatory reports (by country/state) must be supported out-of-the-box?
- Which payment methods are mandatory for phase one (UPI, net banking, cards, wallets)?
- What is the minimum data retention period required by each target market?
- Should some modules (e.g., Social Post, Online Classes) be optional add-ons or core?

9. Release and Phasing Strategy

[Assumption: phased rollout prioritizing operational and financial control]

- **Phase 1** – Organization, Student Info, Admissions, Finance, Attendance, HR, Communications, Reports, Unified Dashboard.
- **Phase 2**– Academics, Exams & Gradebooks, Transport, Hostel, Inventory, Infra Management, Visitor Management, Expense Management.
- **Phase 3**– Assessments, Lesson Delivery, Online Class, Social Post, advanced analytics and BI.

This PRD defines the strategic direction, scope, and detailed requirements for Campus Verse as a unified ERP platform tailored for multi-branch schools, colleges, and universities, enabling owners to manage all institutions from a single, powerful dashboard while preserving the necessary granularity at each branch and academic level.